YX850 AUTOMATIC POWER FAILURE

1. Introduction

The YX850 Automatic Power Failure Switching Module is designed to provide seamless power transition between a primary power source and a backup battery in the event of a power outage. This module is suitable for a wide range of DC applications from 5V to 48V, ensuring continuous operation of critical devices. It is a compact and efficient solution for emergency power conversion.

2. Product Overview

The YX850 module acts as an emergency converter, automatically switching to a connected battery when the main power supply fails. It features a robust relay for switching and clear terminal blocks for easy connection.

Advertisements

2.1. Features

- Automatic power failure detection and switching.
- Wide operating voltage range: DC 5V-48V.
- Maximum load current: 10A.
- Compact design for easy integration.
- Suitable for various electronic projects and emergency power systems.

2.2. Package Contents

- 1 x YX850 Automatic Power Failure Switching Module

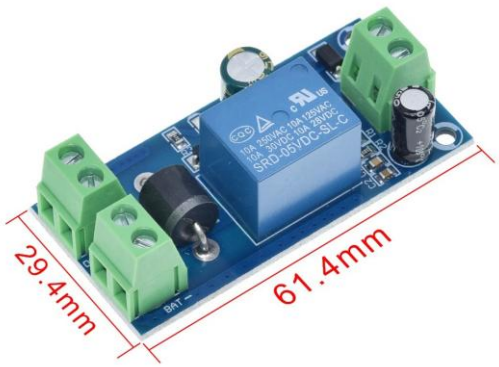


Figure 1: YX850 Module Top View

3. Specifications

Parameter	Value
Product Model	YX850
Operating Voltage Range	DC 5V-48V
Maximum Load Current	10A
Module Type	Mini Electronic Module
Dimensions (L x W x H)	61mm x 30mm x 18mm
Installation Size	23mm (2 x p3MM)
Product Weight	22g
Material	Alloy
Color	Blue

4. Setup and Wiring

Careful wiring is essential for the correct and safe operation of the YX850 module. Please follow the diagram and instructions below.

4.1. Wiring Diagram

Figure 3: YX850 Wiring Diagram

4.2. Connection Steps

1. **Connect Main Power (DC Input)**: Connect your primary DC power adapter or charger to the terminals labeled 'DC +' and 'DC -' on the module. Ensure correct polarity.
2. **Connect Battery (Backup Power)**: Connect your backup battery (e.g., lithium battery) to the terminals labeled 'BAT +' and 'BAT -'. Observe correct polarity.

3. **Connect Load**: Connect your device or load (e.g., LED light, microcontroller) to the 'OUT +' and 'OUT -' terminals. This is where the switched power will be delivered.
4. **Optional: Breaker**: The diagram shows an optional 10A breaker between the battery positive terminal and the module's BAT+ terminal. It is highly recommended to use a suitable fuse or breaker (e.g., 10A) for battery protection.
5. **Power Indicator**: The module has a power indicator (LED) that will illuminate when the main DC power is present.

Important Safety Note: Always ensure correct polarity when connecting power sources and loads to prevent damage to the module and connected devices. Disconnect all power before making or changing connections.

5. Operating Instructions

The YX850 module operates automatically based on the presence of the main DC power supply.

1. **Normal Operation (Main Power Present)**: When the main DC power supply is connected and active, the module will draw power from it and supply it to the connected load. The power indicator LED will be on. The battery will be in a standby state, and if the main power supply also acts as a charger, it will charge the battery (charging functionality is dependent on the external power supply/charger, not the YX850 module itself).
2. **Power Failure (Main Power Absent)**: If the main DC power supply fails or is disconnected, the YX850 module will automatically switch the load's power source from the main DC input to the connected backup battery. This transition is designed to be seamless, minimizing interruption to the load. The power indicator LED will turn off, indicating main power loss.
3. **Power Restoration**: When the main DC power supply is restored, the module will automatically switch the load back to the main power source. The power indicator LED will turn back on.

6. Maintenance

The YX850 module is a low-maintenance device. Follow these guidelines to ensure its longevity:

- Keep the module clean and free from dust and debris.
- Ensure connections are secure and free from corrosion.
- Avoid exposing the module to extreme temperatures, humidity, or direct sunlight.
- Do not exceed the specified maximum load current of 10A.
- Regularly check the health of your backup battery, as its performance directly impacts the module's effectiveness during power outages.

7. Troubleshooting

7.1. No Power to Load

- **Check Main Power**: Verify that the main DC power supply is active and within the 5V-48V range.
- **Check Battery**: Ensure the backup battery is charged and properly connected with correct polarity.
- **Check Connections**: Inspect all wiring for loose connections or incorrect polarity.
- **Check Load**: Ensure the load itself is functioning correctly and not drawing excessive current.
- **Module Damage**: If all connections and power sources are verified, the module might be damaged.

7.2. Module Not Switching Automatically

- **Verify Power Sources**: Ensure both the main DC power and the battery are connected and providing voltage. The module needs to detect the presence/absence of the main power to switch.
- **Relay Functionality**: Listen for a click from the relay when the main power is disconnected/reconnected. If no click, the relay or control circuit might be faulty.

7.3. Overheating

- **Check Load Current**: Ensure the connected load does not exceed the 10A maximum current rating. Excessive current can cause overheating and damage.
- **Ventilation**: Ensure the module has adequate ventilation, especially if enclosed.

8. User Tips

- **Fuse Protection**: Always incorporate a fuse or circuit breaker (e.g., 10A) on the battery line to protect against short circuits and overcurrent, as indicated in the wiring diagram.
- **Battery Type**: While the module is universal, ensure your battery type (e.g., lead-acid, Li-ion) is compatible with your main power supply if it also acts as a charger. The YX850 module itself does not manage battery charging.
- **Voltage Matching**: For optimal performance, try to match the voltage of your main power supply and your backup battery as closely as possible within the 5V-48V range.
- **Testing**: After initial setup, thoroughly test the automatic switching function by intentionally disconnecting and reconnecting the main power supply to ensure your load transitions smoothly.

9. Warranty and Support

For any issues or questions regarding your YX850 Automatic Power Failure Switching Module, please contact your seller or the manufacturer for support. Keep your purchase receipt for warranty claims.